# CHAPTER 3

# [Part 4]

# PROBABILITY THEORY

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# Probability of a General Union of Two Events

If $S$ is any sample space and $A$ and $B$ are any events in $S$, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

# Example 12

In a math class of 30 students, 17 are boys and 13 are girls. On a unit test, 4 boys and 5 girls made an A grade. If a student is chosen at random from the class, what is the probability of choosing a girl or an A student?

**Solution:**




Probabilities: P(girl or A)= P(girl) + P(A) - P(girl and A)

$$= \frac{13}{30} + \frac{9}{30} - \frac{5}{30}$$

$$= \frac{17}{30}$$

# Example 13

Two fair dice are rolled. What is the probability of getting doubles (2 dice showing the same number) or sum of 6?

# Solution

Step 1:
Find the probability of getting double -2 dice showing the same number

- Let A denote the event "get dou
- Doubles can be obtained in 6 w
  (1,1), (2,2), (3,3), (4,4), (5,5), (6,6)
- P(A) = 6/36 = 1/6

# Solution (cont.)

Step 2: Find the probability of getting a sum of 6:

- Let *B* denote the event "get a sum
- Sum of 6 can be obtained in 5 wa
  (1,5), (2,4), (3,3), (4,2), (5,1)

# Solution (cont.)

Step 3: Find the intersection of two events:

- The event $A \cap B$ is "get doubles ar a sum of 6"
- Only 1 way, (3,3)

# Solution (cont.)

Step 4: Find the union of two events

- The probability of getting doubles sum of 6 is,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{6} + \frac{5}{36} - \frac{1}{36} = \frac{5}{18}$$

# Example 14

- Suppose that a student is selecte random among 90 students, whe are over 100 pounds, 20 are boys are over 100 pounds and boys.

- What is the probability that the st selected is over 100 pounds or a

# Solution

- Let A denote the event "the studen
  100 pounds".
  P(A)=35/90

- Let B denote the event "the student
  boy".
  P(A)=20/90

- P(A ∩ B)=15/90

- The probability that the student s
  is over 100 pounds or a boy,

$$P(A \cup B) = \frac{35}{90} + \frac{20}{90} - \frac{15}{90} = \frac{40}{90}$$

# Exercise #2

On New Year's Eve, the probability of a person having a car accident is 0.09. The probability of a person driving while intoxicated is 0.32 and probability of a person having a car accident while intoxicated is 0.15. What is the probability of a person driving while intoxicated or having a car accident?



**Solution:**

- Let A = intoxicated ; B = accident
- Then,

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$
$$= 0.32 + 0.09 - 0.15$$
$$= 0.26$$

# Exercise #3

Mira is going to graduate from a computer science department in a university by the end of the semester. After being interviewed at two companies she likes, she assess that her probability of getting an offer from company A is 0.8, and her probability of getting an offer from company B is 0.6. If she believes that the probability that she will get offers from both companies is 0.5, what is the probability that she will get either from company A or company B (or both)?

# Mutually Exclusive Events

- Two events are mutually exclusive if they cannot occur at the same time.

- Events A and B are mutually exclusive if

$$A \cap B = \emptyset$$

- If A and B are mutually exclusive events,

$$P(A \cup B) = P(A) + P(B)$$

# Mutually Exclusive Events (cont.)

**Mutually Exclusive Events**

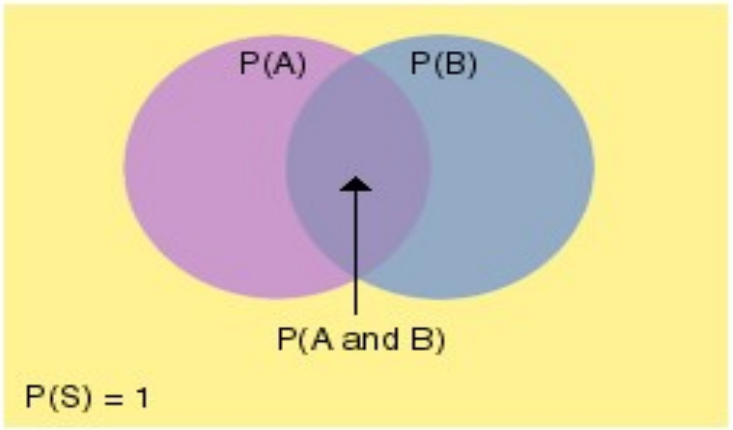
Two events are mutually exclusive if they cannot occur at the same time (i.e., they have no outcomes in common).




In the Venn Diagram above, the probabilities of events A and B are represented by two disjoint sets (i.e., they have no elements in common).

**Non-Mutually Exclusive Events**

Two events are non-mutually exclusive if they have one or more outcomes in common.



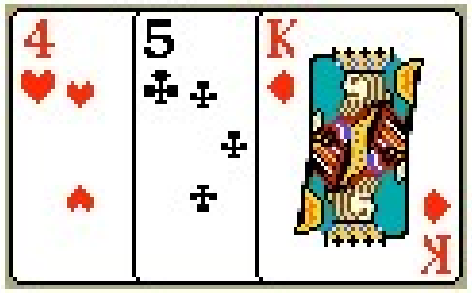

In the Venn Diagram above, the probabilities of events A and B are represented by two intersecting sets (i.e., they have some elements in common).

Note: In each Venn diagram above, the sample space of the experiment is represented by S, with P(S) = 1.

# Example 15

Experiment 1: A single card is chosen at random from a standard deck of 52 playing cards. What is the probability of choosing a 5 or a king?

Possibilities:
1. The card chosen can be a 5.
2. The card chosen can be a king.



Experiment 2: A single card is chosen at random from a standard deck of 52 playing cards. What is the probability of choosing a club or a king?

Possibilities:
1. The card chosen can be a club.
2. The card chosen can be a king.
3. The card chosen can be a king and a club (i.e., the king of clubs).

# Example 16

**Question:** Two fair dice are rolled. Find the probability of getting doubles or the sum of 5?

**Solution:**

- Let A denote the event "get doubles"
- Let B denote the event "get the sum of 5"
- A and B are mutually exclusive (you cannot get doubles and the sum of 5 simultaneously)

- A ={(1,1) (2,2) (3,3) (4,4) (5,5) (
  $P(A) = 6/36 = 1/6$
- B ={(1,4) (2,3) (3,2) (4,1)



- The probability of getting doubles or the sum of 5 is,
  $P(A \cup B) = 1/6 + 1/9 = 5/18$

# Example 17

A glass jar contains 1 **RED**, 3 **GREEN**, 2 **BLUE** and 4 **YELLOW** marbles. If a single marble is chosen at random from the jar, what is the probability that it is **YELLOW** or **GREEN** ?

**Solution:**

Probabilities:

$$P(\text{yellow}) = \frac{4}{10}$$

$$P(\text{green}) = \frac{3}{10}$$

$$P(\text{yellow or green}) = P(\text{yellow}) + P(\text{green})$$

$$= \frac{4}{10} + \frac{3}{10}$$

$$= \frac{7}{10}$$

